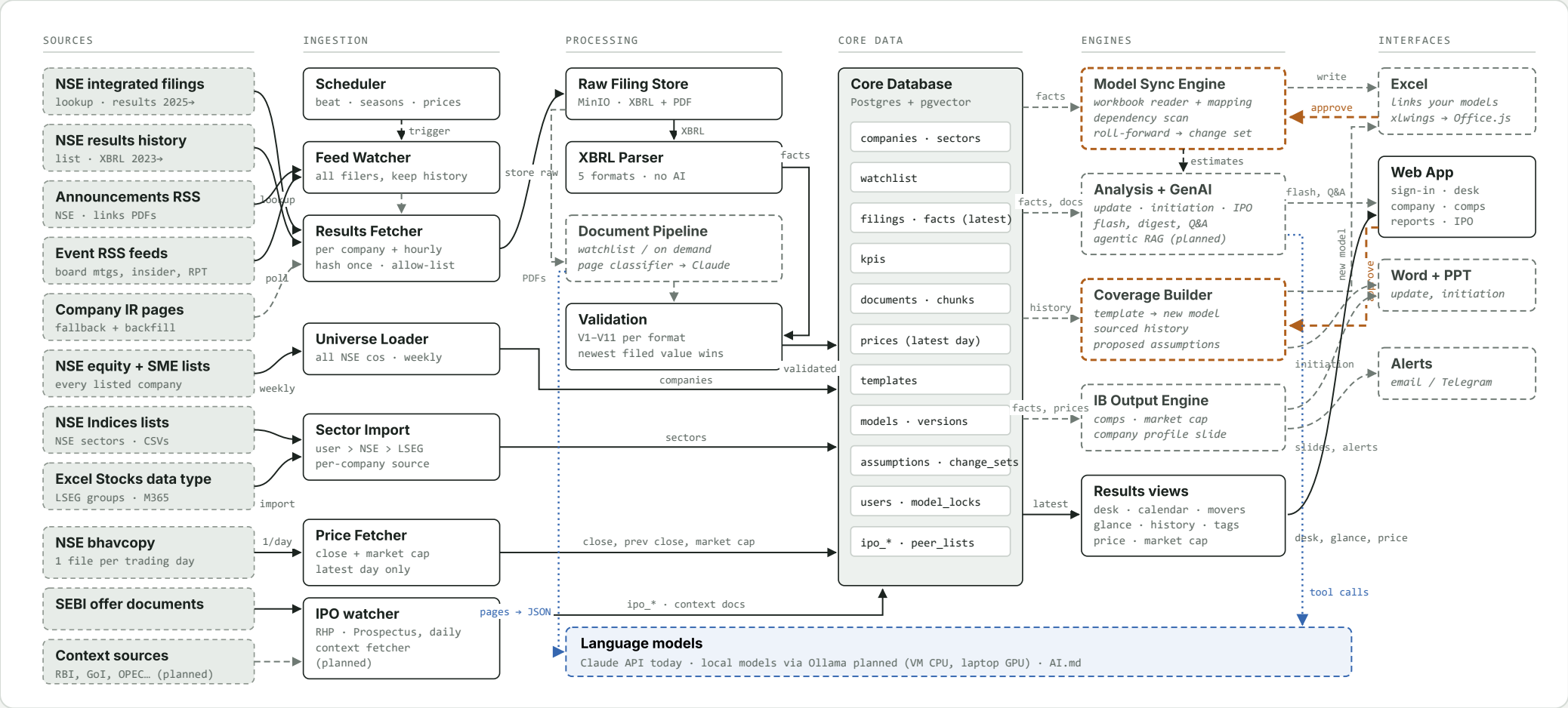
Analekh Architecture

The analyst's record, kept for you.

Every NSE-listed company is preloaded. Its results are fetched from NSE's integrated filings and, back to March 2023, its financial-results list, then parsed and validated on both bases for every filer. Closing prices come once a day from NSE's bhavcopy, board meetings fill a results calendar, and SEBI's RHP and Prospectus lists start coverage of recent listings. Signed-in analysts update their Excel models after approving each change, or build one from a sector template, and draft reports in Word and PowerPoint. Core numbers come from XBRL; the AI writes the analysis and pulls every figure from the database. Proposed: an agent that reads official and research sources fetched ahead, questions the analyst's estimates, and learns each firm's and analyst's voice.

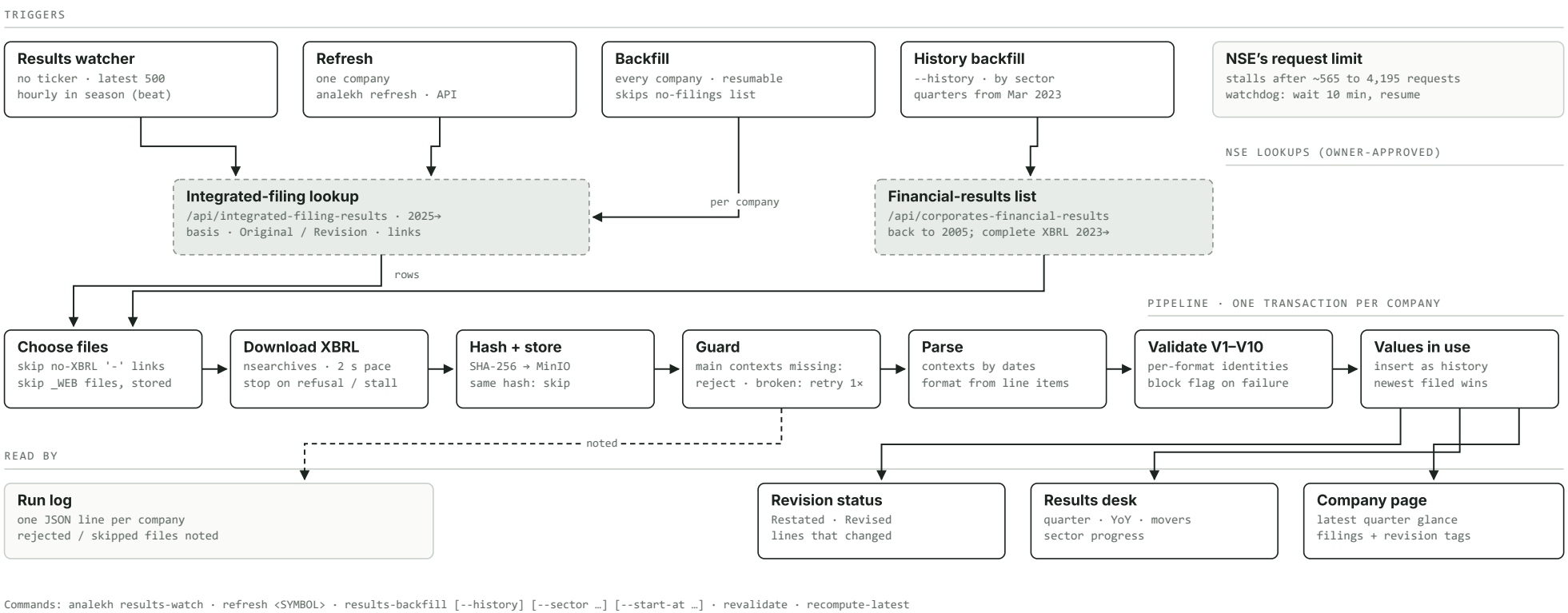
Components and data flow



data · control / return · analyst approval · Claude API call · external source · built, running on real data · core built or planned, not running yet

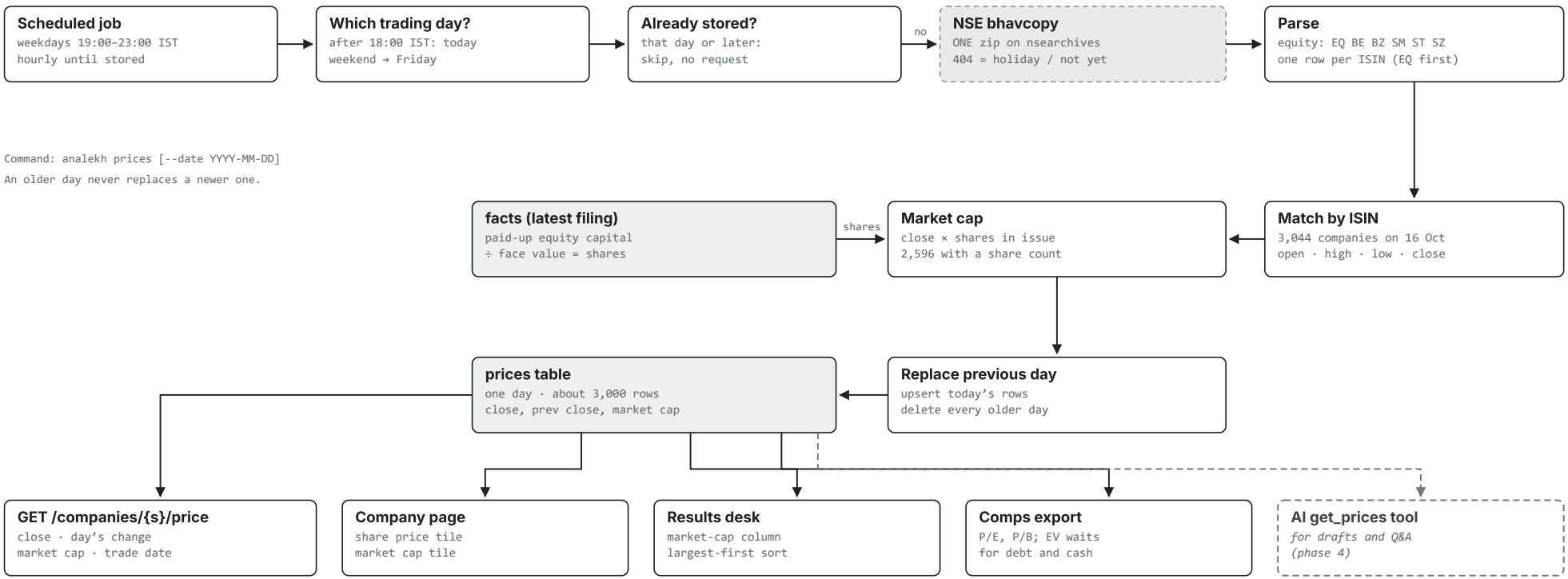
Results come from NSE's integrated-filing lookup (per company, and one request every hour in season without a ticker for the latest filings) and its financial-results list for 2023–2024. When a quarter is refiled, the most recently filed values are used and the older ones kept as history. Sectors follow NSE's own taxonomy first. Prices: one bhavcopy per trading day; only the latest close is kept. Celery beat runs every job in Docker. Only the relevant PDF pages will reach Claude, and nothing is written to Excel until the analyst approves.

Workflow 1 · Results collection BUILT



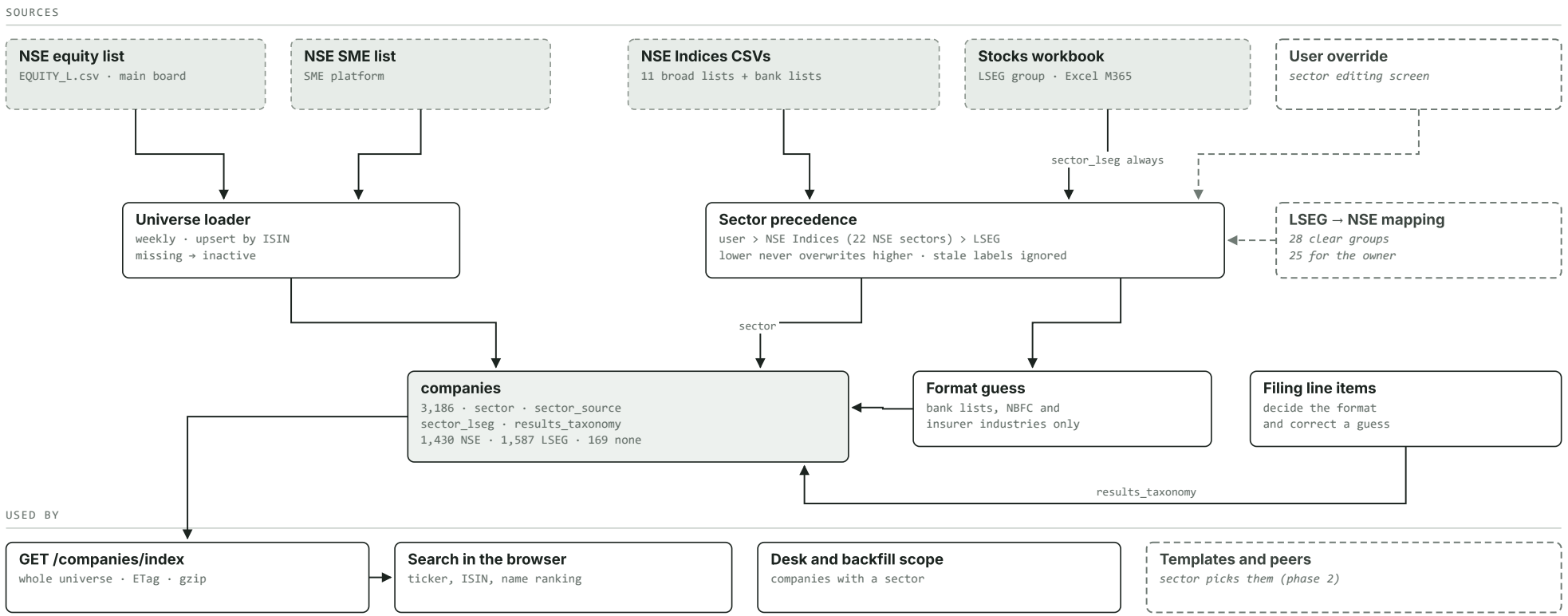
Four triggers, two NSE lookups, one pipeline. The watcher asks once, without a ticker, for the latest filings across all companies; a refresh or backfill asks per company. The history backfill reads the financial-results list, whose files are complete from March 2023. Each file is downloaded once (SHA-256, stored in MinIO); files missing their main contexts are rejected and a broken file is retried once, then skipped, so one bad file never costs a company its other filings. Values are inserted as history and the newest filed value of each line becomes the one in use, whatever order files arrived in. One transaction per company: a stopped run keeps what it stored and resumes at the company it stopped on. NSE pauses after a few hundred to a few thousand requests; runs wait 10 minutes and go on.

Workflow 2 · Daily closing prices BUILT



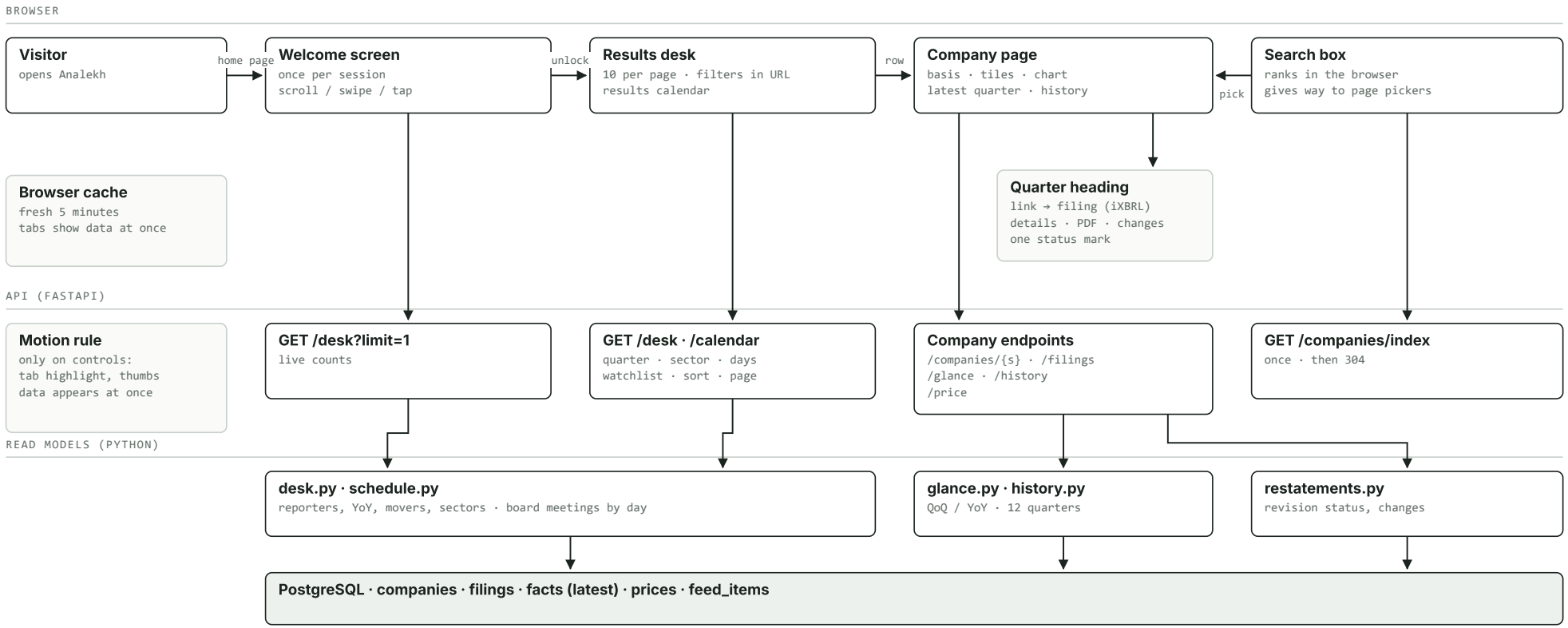
One file per trading day, after the close. NSE publishes the bhavcopy in the evening; the job tries hourly from 19:00 to 23:00 IST on weekdays until the day is stored, then does nothing. Market cap is the close times shares in issue, worked out from the latest results filing (paid-up equity capital divided by face value). Only the latest close is kept (owner's rule, 18 October 2026): each file replaces the previous day's prices, and the day's change comes from the previous close the file carries. No live prices, no price history.

Workflow 3 · Company list and sectors BUILT



Every NSE company, then its sector. The equity and SME lists are loaded weekly. Sectors follow NSE's own taxonomy (22 sectors) from NSE Indices' published constituent lists, which outrank the LSEG industry groups imported from an Excel Stocks data type workbook; a user's value will outrank both. The LSEG group is always kept in its own column. The results format (Ind-AS, bank, NBFC, general or life insurer) is first guessed from strong signals only, then decided by the line items each filing tags.

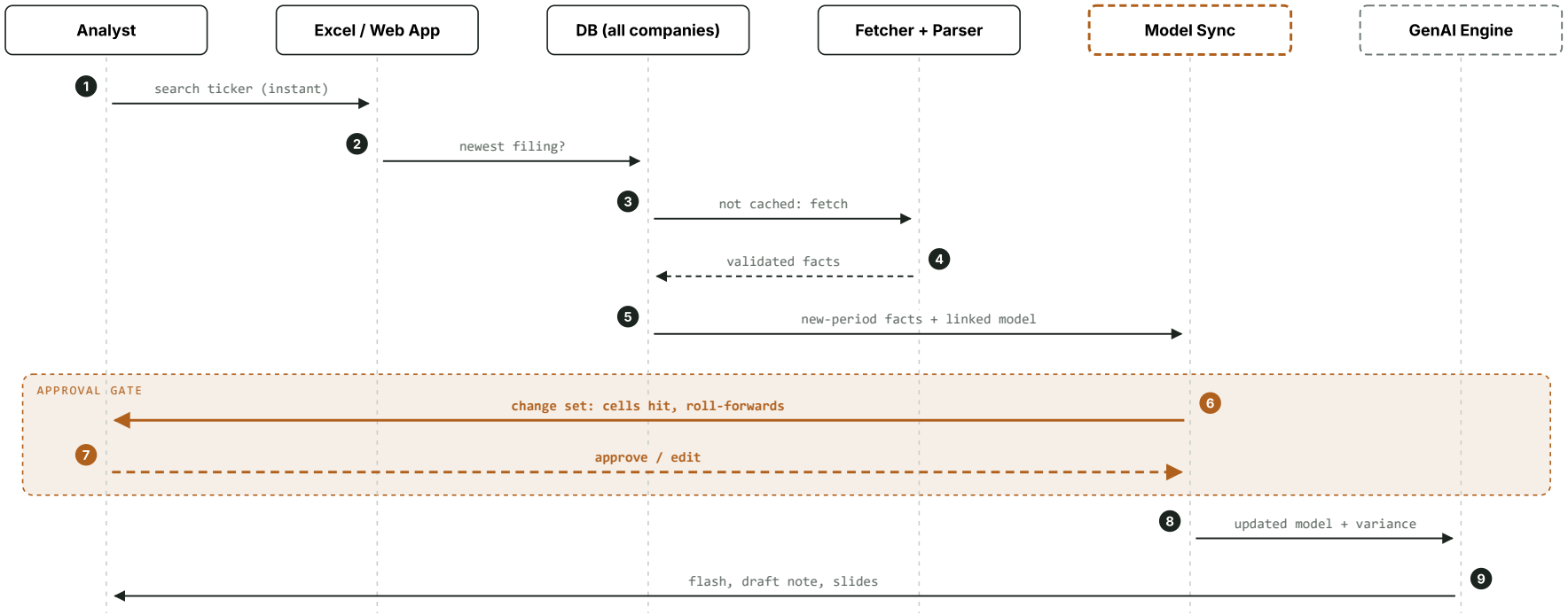
Workflow 4 · Opening Analekh and reading results BUILT



Welcome, then the desk. The first visit of a browser session to the home page shows the welcome screen with live counts; a scroll, swipe or tap unlocks it into the Results desk and its results calendar. Search runs in the browser over the whole universe. On a company page, the results history shares one request with the tiles and the chart; each quarter's heading links its filing on NSE (iXBRL, the newest after a refiling) and opens its details. Data stays fresh in the browser for five minutes, so moving between tabs shows it at once. Motion is only on what you click; figures appear immediately.

Workflow 5 · Quarterly model update

PLANNED · PHASE 2



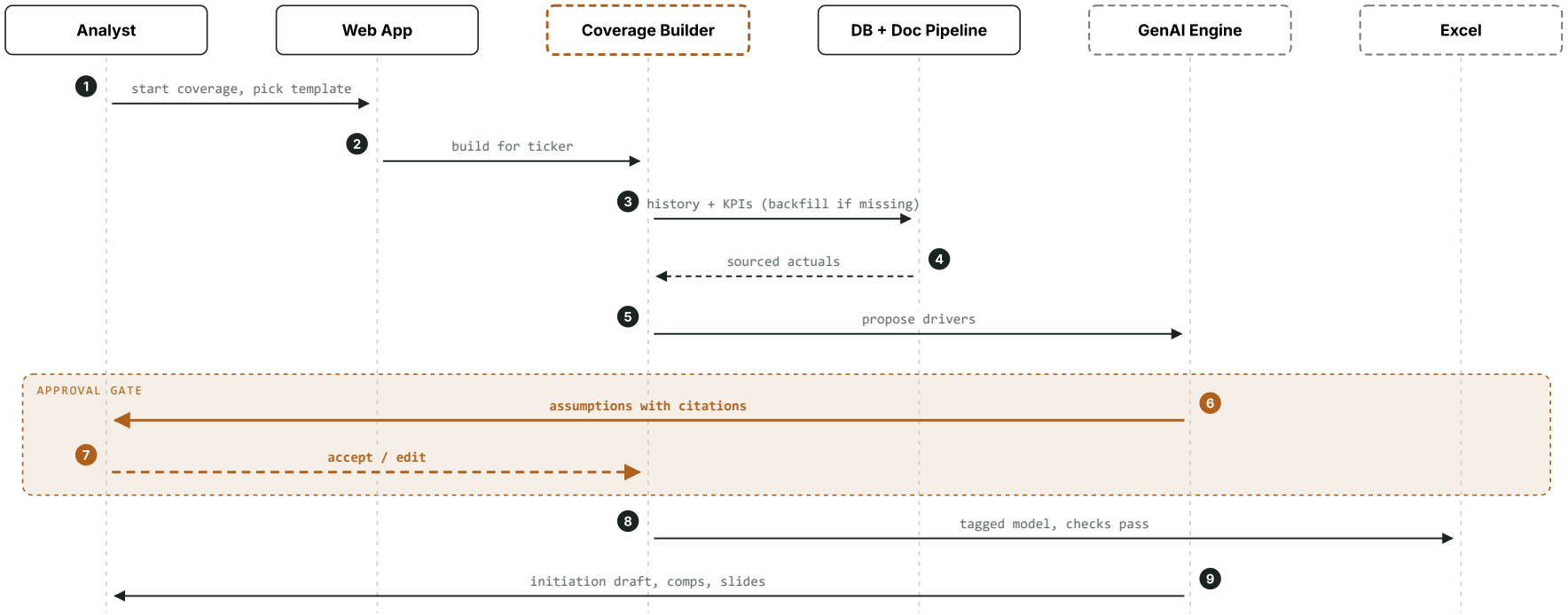
Steps 3 and 4 run only when the filing isn't already in the database (for most companies it is, collected by workflow 1). The linked model is found through watched folders or the open workbook, identified by a hidden tag, its file name or its contents. Steps 6 and 7 are the approval gate: the analyst sees which cells change and which forecasts need rolling forward, accepts or edits each one, and only then is a new model version saved. The original workbook is never overwritten.

What changed in v0.0.6

- **Sign-in and roles:** every route signed in; admin and analyst; first password and reset codes by email; audit fixes in.
- **Model locks:** one holder per model; others read-only until it frees.
- **Recalc:** LibreOffice in its own container on an internal network.
- **Hosting:** Oracle Cloud Mumbai VM with CI runners and nightly backups.
- **IPO coverage:** SEBI's RHP and Prospectus to restated figures and section drafts.
- **Proposed:** agentic RAG over official sources, conflict flags on estimates, local models, learning each analyst's voice.

Workflow 6 · Start coverage

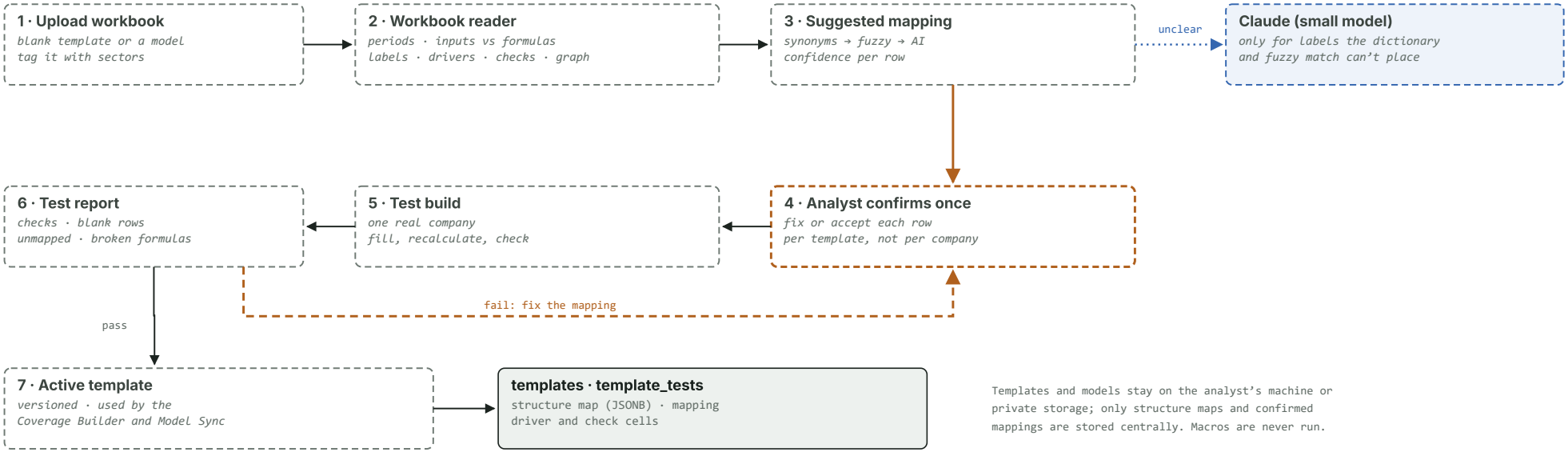
PLANNED · PHASE 2



Used when no model exists yet. The analyst picks a company and a sector template (their own upload or a built-in one). The builder fills the template's inputs with sourced history and keeps its formulas. Proposed assumptions pass through the same approval gate before the tagged model is saved. From the next results onward, the quarterly update takes over.

Workflow 7 · Template onboarding

PLANNED · PHASE 2

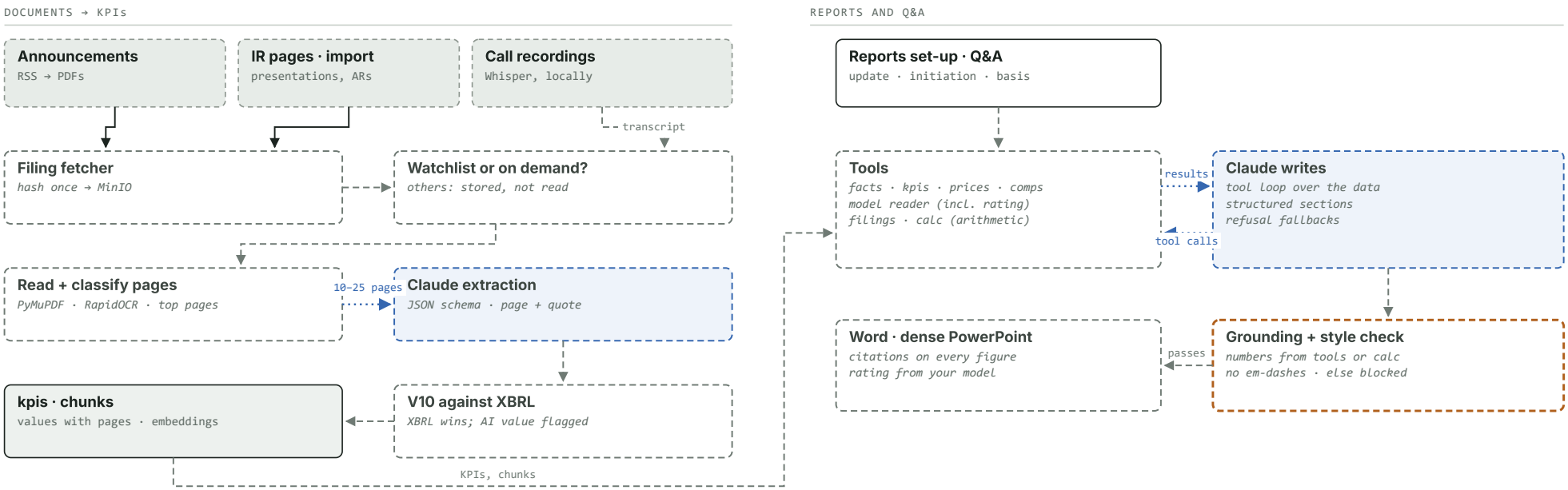


Confirm once per template. The workbook understanding engine reads any workbook: its period axis, inputs versus formulas, row labels, drivers, checks and the formula graph. It suggests where each standard line goes, with a confidence for each row; the analyst confirms or fixes it once. A test build with a real company must pass (checks hold, no broken formulas) before the template can be used.

Workflow 8 · Documents, KPIs and reports

REPORTS SET-UP BUILT

DRAFTING · PHASES 3–4 · LIVE CALLS: FUTURE SCOPE



Built: the Reports set-up on real data; page classifier, extraction schema, transcript parser, local call transcription, tools, calc, grounding check and the report loop (synthetic tests).

Next: the Word and PowerPoint writers and the style check. Waiting on: a valid API key, a presentation and transcript source, the embeddings provider.



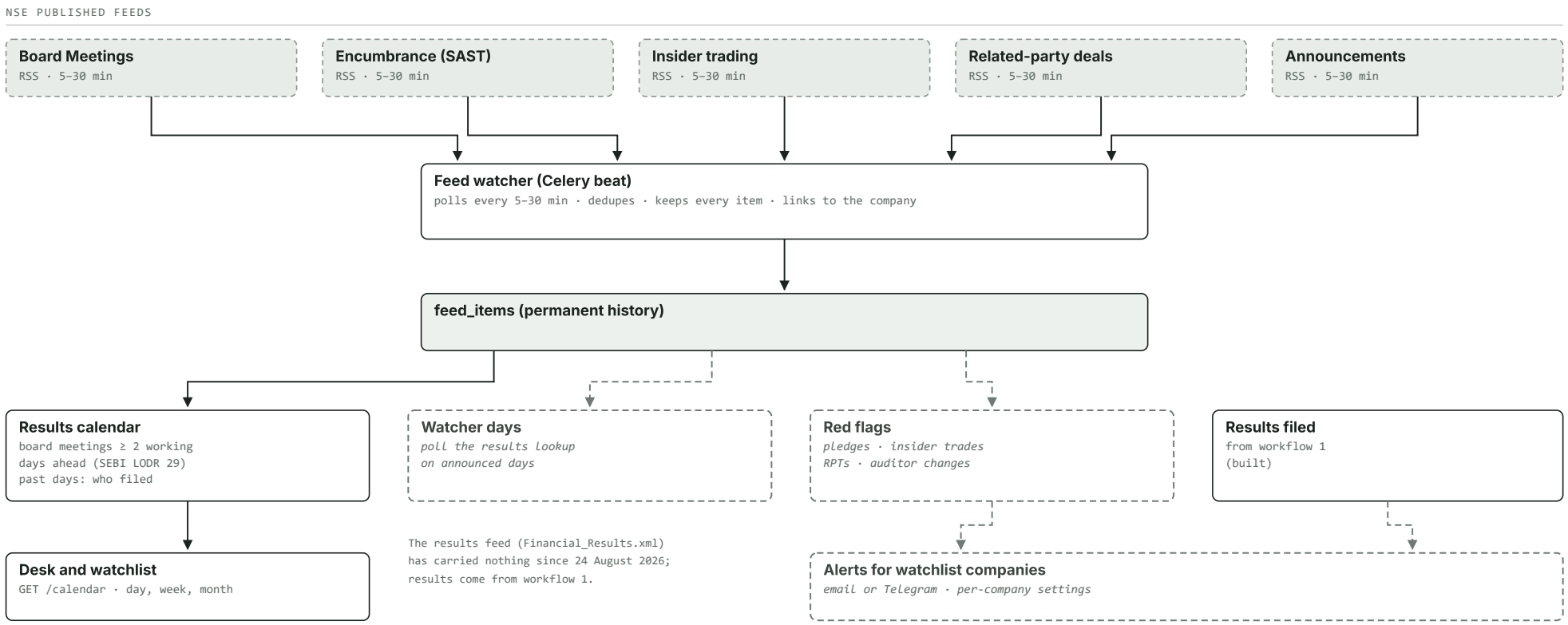
Needs an always-on machine during the call (section 22.7); phone-only calls would need a paid telephony service. Transcripts stay in the analyst's own database; XBRL wins on any figure.

Only selected pages reach Claude, and every number is checked. For watchlist or on-demand companies, PDFs are read locally, pages classified and only the relevant ones sent for extraction against a JSON schema, each value with its page and quote. XBRL wins over any AI value. The analyst sets up a quarterly update or an initiating coverage note on the Reports screen (company, basis, quarter; built). Drafts are written in a tool loop over the database, the comps and the model; a grounding check blocks any number that did not come from a tool result or a calculation, and a style check blocks em-dashes and other marks of generated text. The rating comes from the analyst's model or the analyst, never from Analekh. Future scope: a listener that transcribes the earnings call live.

Workflow 9 · Events, results calendar and alerts

FEEDS · CALENDAR BUILT

RED FLAGS · ALERTS · PHASE 5

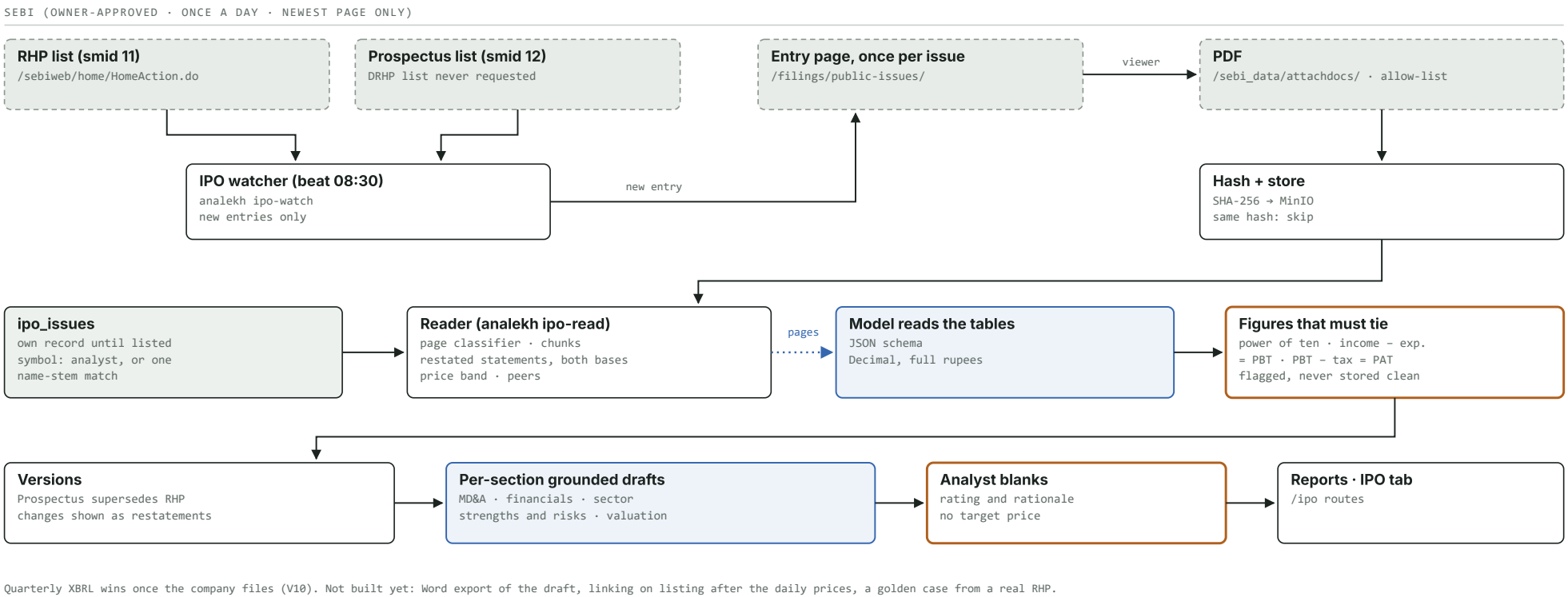


NSE's published event feeds. The feed watcher runs from Celery beat and keeps every item. Companies must announce a results board meeting at least two working days ahead (SEBI LODR 29), so the Board Meetings feed now builds the results calendar on the desk and the watchlist; a past day shows who actually filed. Next: telling the results watcher which days matter, and alerts for watchlist companies when they file results or when a red flag appears (pledges, insider trades, related-party deals, auditor changes).

Workflow 10 · IPO coverage from offer documents

BUILT · SYNTHETIC TESTS

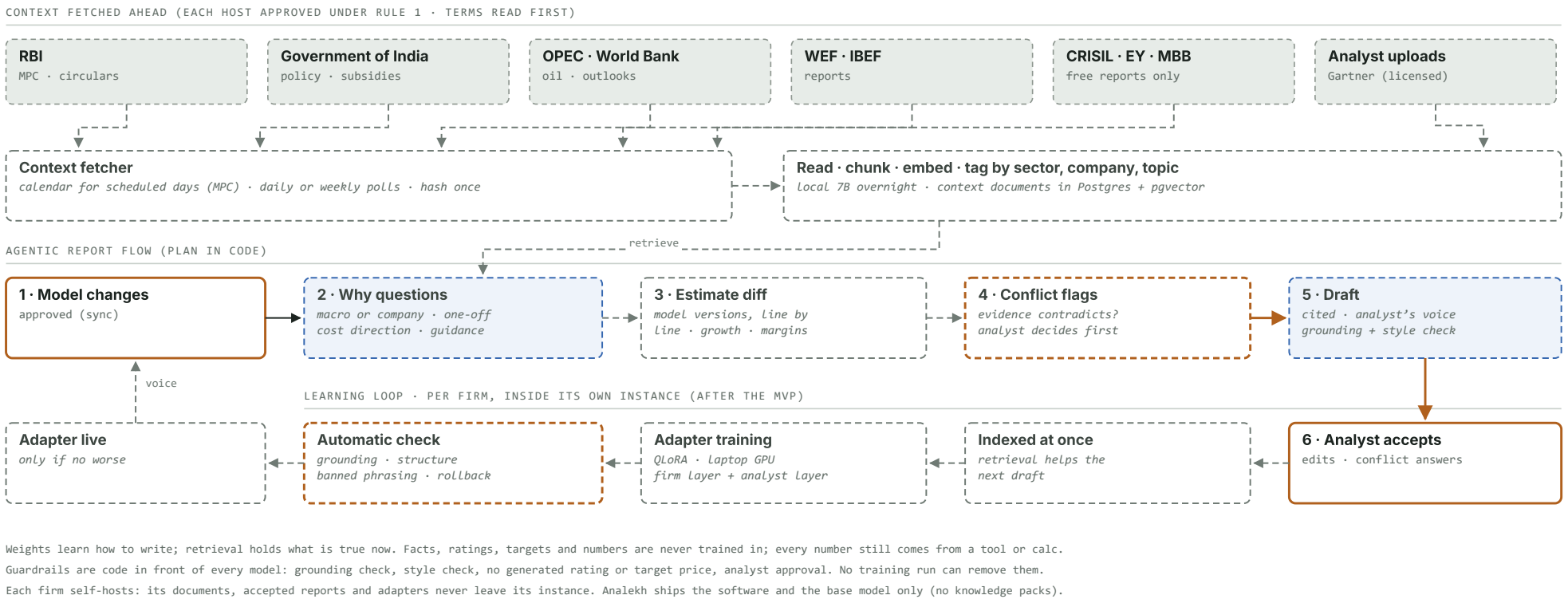
FIRST REAL RHP RUN PENDING



Coverage before the first quarterly result. Once a day the watcher reads the newest page of SEBI's RHP and Prospectus lists only (owner-approved host and paths; the DRHP list is never requested), fetches each new entry page once and takes the PDF from its embedded viewer after checking it against the allow-list. An issue is its own record until NSE assigns a symbol. The reader finds the restated statements, price band and peers; figures that do not tie are flagged, never stored clean. The Prospectus supersedes the RHP and changes show as restatements; quarterly XBRL wins once filed. Drafts run one grounded loop per section; the rating and rationale are left for the analyst, with no target price.

Workflow 11 · Agentic reports and learning

PROPOSED · V0.0.6 · AI.MD



Weights learn how to write; retrieval holds what is true now. Facts, ratings, targets and numbers are never trained in; every number still comes from a tool or calc.

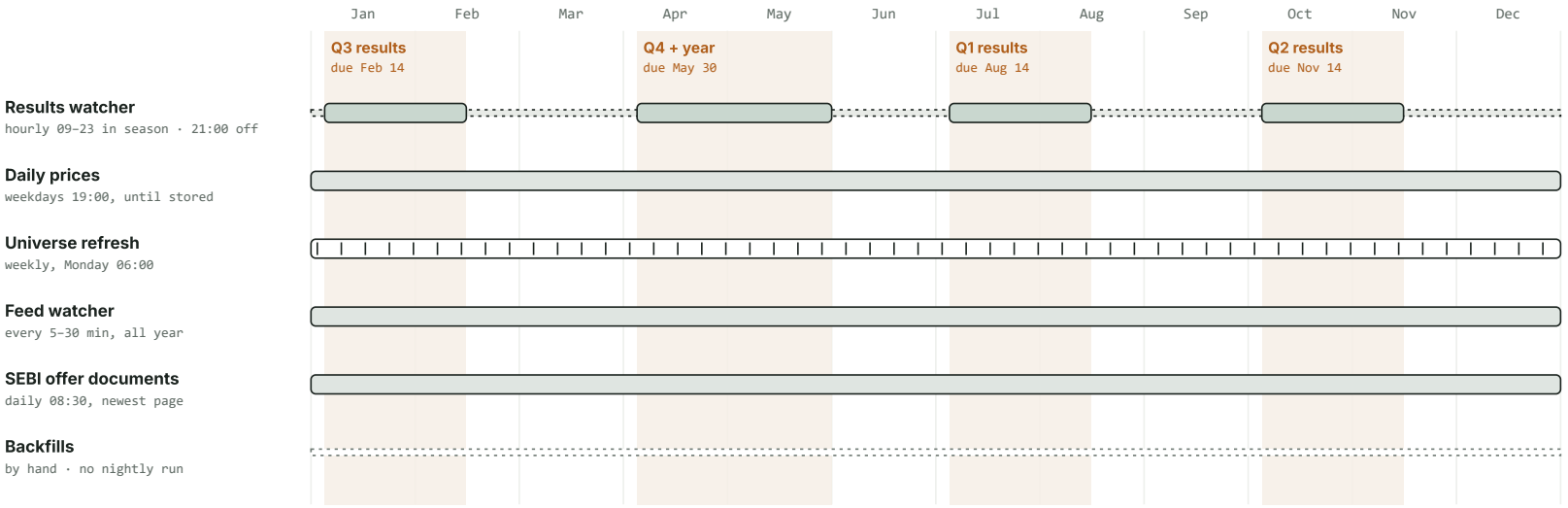
Guardrails are code in front of every model: grounding check, style check, no generated rating or target price, analyst approval. No training run can remove them.

Each firm self-hosts: its documents, accepted reports and adapters never leave its instance. Analekh ships the software and the base model only (no knowledge packs).

Numbers are not the only evidence. Official and research sources are fetched as they publish (by calendar for scheduled events such as RBI's MPC, by polling otherwise) and tagged by sector, so the agent only retrieves. It asks standing "why" questions of every movement, diffs the analyst's estimate changes and flags any the evidence contradicts before drafting; the analyst decides, the agent never changes an estimate. Accepted reports and edits train small adapters per firm and per analyst, swapped in only after an automatic check. Each firm hosts its own instance; its data never leaves it.

When each job runs

BUILT · CELERY BEAT ON THE VM

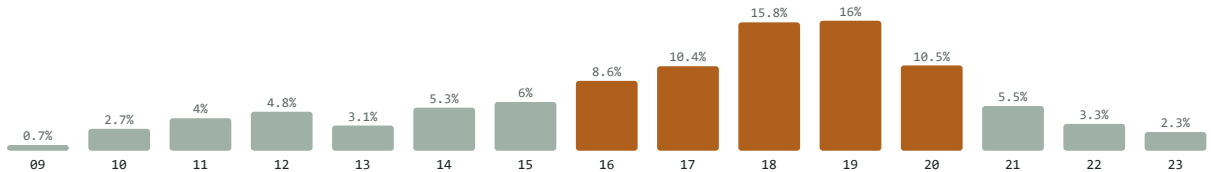


Requests a year for the results watcher: about 3,000 by season (it was about 5,500 hourly all year). Windows: day 5 to two days past each deadline.

When results are filed (IST)

share of filings by hour; 00–09 h: 1.0%

16–20 h: 61%

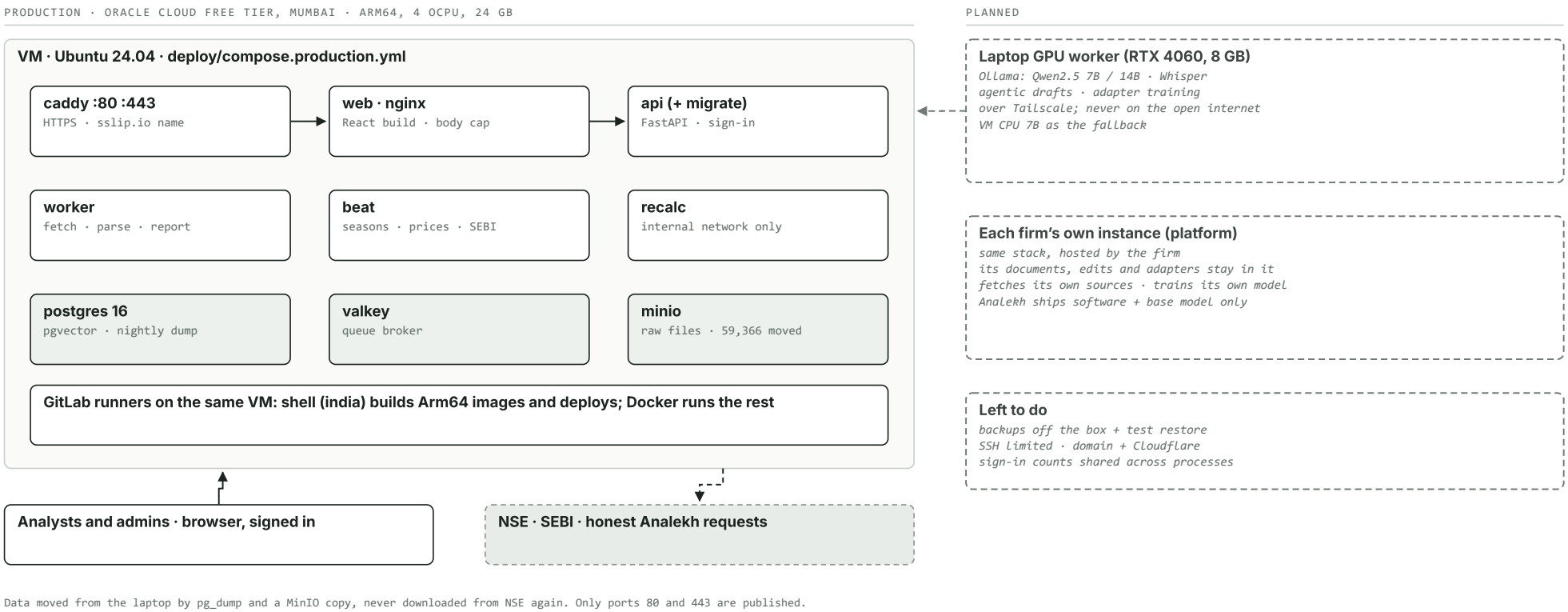


Results arrive in four seasons. SEBI requires quarterly results within 45 days of the quarter (60 for the year end). NSE showed integrated filings a median of one second after the company filed (299 filings, 24 companies), so any delay is the watcher's interval. Since 2023, every first filing of those companies fell between day 8 and day 59, inside the windows; under 1% were filed between midnight and 09:00. The watcher polls every hour from 09:00 to 23:00 in season and once a day at 21:00 otherwise: about 45% fewer requests than every hour all year.

Where it runs

ORACLE CLOUD MUMBAI · SINCE 26 OCT

GPU WORKER · PER-FIRM INSTANCES



Data moved from the laptop by pg_dump and a MinIO copy, never downloaded from NSE again. Only ports 80 and 443 are published.

Numbers come from data

Figures in drafts are fetched from the database or the model through tool calls, and each one carries its XBRL tag or PDF page reference.

Data rights

NSE's published feeds and lists, two lookups (owner-approved; per company, plus one request per watcher run), one bhavcopy a day, NSE Indices' CSVs, SEBI's RHP and Prospectus lists. Requests identify as Analekh, never as a browser. Each new source needs the owner's approval.

Stack

Python 3.12 · FastAPI · Celery + Valkey · lxml · PyMuPDF · RapidOCR · Claude API → Ollama · Postgres + pgvector · MinIO · LibreOffice (recalc) · React + pnpm · Caddy · Docker Compose · GitLab CI